# GO_Biological_Process_2023 Top pathways by non−permulation

| Geneset | stat | num.genes | pval | p.adj | gene.vals |
|---|---|---|---|---|---|
| tRNA Modification (GO:0006400) | 0.19176247 | 66 | 7.358e−08 | 3.965e−04 | METTL2B:155 THADA:190 PUS10:237 TRMT10A:259 DTWD1:305 FTSJ1:343 |
| tRNA Methylation (GO:0030488) | 0.24706383 | 36 | 2.936e−07 | 7.909e−04 | METTL2B:155 THADA:190 TRMT10B:258 TRMT10A:259 FTSJ1:343 METTL8:374 |
| Fatty Acid Catabolic Process (GO:0009062 | 0.19125506 | 58 | 4.810e−07 | 8.639e−04 | ECI2:10 ACAD11:177 SLC27A4:245 EHHADH:348 ACOT8:398 HACL1:516 |
| Chemical Synaptic Transmission (GO:00072 | −0.09064213 | 254 | 7.310e−07 | 9.846e−04 | NPTX1:32 RIMBP2:42 GABRR1:52 RPS6KA2:67 NTSR1:129 GABRB3:145 |
| Muscle Contraction (GO:0006936) | −0.15221231 | 87 | 9.545e−07 | 1.029e−03 | TACR2:22 MYH2:100 KCNJ12:182 LMOD2:252 MYH8:261 MYH1:341 |
| DNA Repair (GO:0006281) | 0.07865785 | 273 | 8.488e−06 | 4.739e−03 | FANCM:50 FANCE:108 RHNO1:112 MBD4:124 MSH3:225 PARP4:284 |
| RNA Methylation (GO:0001510) | 0.17155860 | 57 | 7.590e−06 | 4.739e−03 | METTL4:115 METTL2B:155 THADA:190 TRMT10A:259 FTSJ1:343 METTL8:374 |
| RNA Modification (GO:0009451) | 0.19039804 | 47 | 6.391e−06 | 4.739e−03 | METTL2B:155 DTWD1:305 FTSJ1:343 TRMT1:713 PUS3:834 PUS7L:954 |
| Regulation Of Synaptic Transmission, Glu | −0.17499561 | 54 | 8.795e−06 | 4.739e−03 | TSHZ3:95 TPRG1L:204 GRM6:268 DGKI:536 SHANK3:638 DRD2:654 |
| Visual Perception (GO:0007601) | −0.13421604 | 95 | 6.355e−06 | 4.739e−03 | TRPM1:17 VAX2:19 GNAT2:25 CRYAA:118 CABP2:122 OAT:133 |
| Calcium Ion Import Across Plasma Membran | −0.22244737 | 33 | 9.842e−06 | 4.821e−03 | TRPM1:17 SCN1A:219 CACNA1B:265 SCN2A:338 CLU:761 SLC8A3:855 |
| Double−Strand Break Repair (GO:0006302) | 0.10187575 | 157 | 1.113e−05 | 4.999e−03 | HELQ:313 MCM9:315 TP53BP1:342 RNF168:352 DNA2:380 AUNIP:570 |
| Striated Muscle Contraction (GO:0006941) | −0.17272792 | 52 | 1.665e−05 | 6.903e−03 | MYH7B:132 MYH8:261 PGAM2:355 RYR2:386 TNNT3:391 KCNQ1:423 |
| Double−Strand Break Repair Via Homologou | 0.12081336 | 105 | 1.953e−05 | 7.014e−03 | FANCM:50 HELQ:313 MCM9:315 XRCC1:337 AUNIP:570 ERCC5:638 |
| Fatty Acid Beta−Oxidation (GO:0006635) | 0.18069223 | 47 | 1.844e−05 | 7.014e−03 | ECI2:10 ACOX2:63 ACAD11:177 GCDH:261 EHHADH:348 ACOT8:398 |
| Regulation Of Chromosome Separation (GO: | 0.40811282 | 9 | 2.240e−05 | 7.543e−03 | NCAPD2:110 SMC4:118 SMC2:883 NUMA1:1002 NCAPG:1278 TPR:1904 |
| Modulation Of Chemical Synaptic Transmis | −0.11320509 | 117 | 2.424e−05 | 7.684e−03 | NPTX1:32 TPRG1L:204 CLSTN1:256 CACNA1B:265 GRM6:268 CPLX3:408 |
| tRNA Processing (GO:0008033) | 0.17363651 | 49 | 2.645e−05 | 7.918e−03 | METTL2B:155 DTWD1:305 FTSJ1:343 ADAT1:705 TRMT1:713 PUS3:834 |
| Mitochondrial Gene Expression (GO:014005 | 0.12019508 | 100 | 3.371e−05 | 9.560e−03 | PTCD3:246 LARS2:358 FASTKD1:446 MRPS10:475 TFB2M:498 MRPS9:696 |
| Mitochondrial Translation (GO:0032543) | 0.12088420 | 96 | 4.370e−05 | 1.177e−02 | PTCD3:246 MTIF2:317 LARS2:358 MRPS10:475 TSFM:606 GFM1:695 |
| Fatty Acid Oxidation (GO:0019395) | 0.16547344 | 47 | 8.773e−05 | 2.251e−02 | ECI2:10 ACAD11:177 EHHADH:348 HACL1:516 ACADM:530 HAO2:573 |
| Cell Junction Assembly (GO:0034329) | −0.12458697 | 82 | 9.814e−05 | 2.403e−02 | SHANK2:13 DBNL:81 SDK1:85 CDH26:277 CDH6:373 ITGA6:405 |
| Regulation Of Heart Rate By Cardiac Cond | −0.18148755 | 38 | 1.091e−04 | 2.555e−02 | BIN1:272 DSG2:400 KCNQ1:423 CAV1:468 CAMK2D:639 DSC2:1099 |
| Non−Motile Cilium Assembly (GO:1905515) | 0.19944969 | 31 | 1.221e−04 | 2.741e−02 | ARL13B:55 CEP89:262 IFT122:403 C2CD3:433 TOGARAM1:539 TBC1D32:627 |
| Actin−Myosin Filament Sliding (GO:003327 | −0.33279247 | 11 | 1.326e−04 | 2.804e−02 | MYH2:100 MYH8:261 MYH7:580 TNNT2:672 MYL1:1074 TPM1:1112 |
| Alpha−Amino Acid Catabolic Process (GO:1 | 0.20476280 | 29 | 1.361e−04 | 2.804e−02 | TAT:217 SARDH:303 ALDH4A1:339 KMO:340 HIBCH:752 HMGCLL1:1047 |
| Regulation Of Chromosome Segregation (GO | 0.22939119 | 23 | 1.405e−04 | 2.804e−02 | NCAPD2:110 SMC4:118 SMC2:883 CDCA2:983 NUMA1:1002 NCAPG:1278 |
| Anterograde Trans−Synaptic Signaling (GO | −0.08117521 | 181 | 1.730e−04 | 3.214e−02 | NPTX1:32 GABRR1:52 RPS6KA2:67 NTSR1:129 GABRB3:145 SLC18A2:159 |
| Recombinational Repair (GO:0000725) | 0.11548165 | 89 | 1.697e−04 | 3.214e−02 | RHNO1:112 HELQ:313 MCM9:315 AUNIP:570 ERCC5:638 RAD51AP1:644 |
| Cardiac Conduction (GO:0061337) | −0.16373310 | 43 | 2.050e−04 | 3.563e−02 | BIN1:272 DSG2:400 KCNQ1:423 CAV1:468 CAMK2D:639 SLC9A1:900 |
| Sister Chromatid Segregation (GO:0000819 | 0.18690837 | 33 | 2.037e−04 | 3.563e−02 | CHAMP1:73 SMC4:118 SPAG5:334 ESPL1:356 KIFC1:401 KIF18B:468 |
| Monocarboxylic Acid Catabolic Process (G | 0.23780537 | 20 | 2.323e−04 | 3.826e−02 | CYP26C1:147 ACOT8:398 HOGA1:1129 LPIN3:1469 ACOX1:1805 FAAH:1880 |
| Sensory Perception Of Light Stimulus (GO | −0.10880963 | 96 | 2.344e−04 | 3.826e−02 | TRPM1:17 VAX2:19 GNAT2:25 CRYAA:118 CABP2:122 OAT:133 |
| NADH Dehydrogenase Complex Assembly (GO: | 0.15189251 | 48 | 2.745e−04 | 4.225e−02 | ECSIT:87 NDUFAF2:173 NDUFS3:424 TMEM126A:742 NDUFAF1:941 NDUFB8:1065 |
| Mitochondrial Respiratory Chain Complex | 0.15189251 | 48 | 2.745e−04 | 4.225e−02 | ECSIT:87 NDUFAF2:173 NDUFS3:424 TMEM126A:742 NDUFAF1:941 NDUFB8:1065 |
| Positive Regulation Of Chromosome Separa | 0.31357597 | 11 | 3.170e−04 | 4.745e−02 | NCAPD2:110 SMC4:118 SMC2:883 NUMA1:1002 NCAPG:1278 NCAPH:2371 |
| Maturation Of SSU−rRNA (GO:0030490) | 0.18172002 | 32 | 3.762e−04 | 5.478e−02 | RPS16:467 TSR2:543 RRP36:654 SRFBP1:769 TSR1:953 WDR46:1026 |
| DNA Metabolic Process (GO:0006259) | 0.06255285 | 274 | 3.894e−04 | 5.521e−02 | PARG:46 MBD4:124 MSH3:225 PARP4:284 MCM9:315 WDHD1:330 |
| Cardiac Muscle Contraction (GO:0060048) | −0.18594522 | 30 | 4.256e−04 | 5.880e−02 | RYR2:386 KCNQ1:423 MYH7:580 CAMK2D:639 TNNT2:672 GRK2:946 |
| Maturation Of SSU−rRNA From Tricistronic | 0.20266026 | 25 | 4.540e−04 | 5.966e−02 | UTP20:93 RPS16:467 TSR2:543 RRP36:654 KRI1:817 TSR1:953 |

# EnrichmentHsSymbolsFile2 Top pathways by non−permulation

| Geneset | stat | num.genes | pval | p.adj | gene.vals |
|---|---|---|---|---|---|
| HAMAI_APOPTOSIS_VIA_TRAIL_UP | 0.08804919 | 591 | 3.507e−13 | 2.276e−09 | SPG11:2 AGGF1:5 PJA2:7 ENPP2:30 KIF16B:64 CEP192:65 |
| REACTOME_NEURONAL_SYSTEM | −0.09845701 | 382 | 4.581e−11 | 1.486e−07 | ADCY1:8 CAMK2B:11 SHANK2:13 GABRR1:52 RPS6KA2:67 DBNL:81 |
| REACTOME_MUSCLE_CONTRACTION | −0.13868862 | 185 | 8.270e−11 | 1.788e−07 | CAMK2B:11 ITPR1:66 KCNJ12:182 NEB:190 MYBPC2:194 SCN1A:219 |
| NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO | −0.15450244 | 127 | 1.892e−09 | 3.069e−06 | NPBWR2:28 CABLES2:69 ZBTB46:105 NTSR1:129 MYT1:146 TCFL5:147 |
| MARTENS_TRETINOIN_RESPONSE_UP | −0.06750341 | 675 | 2.805e−09 | 3.418e−06 | MMEL1:5 SHANK2:13 TRIM29:18 CHST15:23 NPTX1:32 KLK7:35 |
| REACTOME_G_ALPHA_I_SIGNALLING_EVENTS | −0.10554753 | 267 | 3.161e−09 | 3.418e−06 | ADCY1:8 CAMK2B:11 GNAT2:25 NPBWR2:28 OPN1LW:63 ITPR1:66 |
| FISCHER_DREAM_TARGETS | 0.05805156 | 873 | 7.771e−09 | 7.203e−06 | MDM1:27 MTBP:32 FANCM:50 ARL13B:55 CEP192:65 FANCE:108 |
| MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M | −0.09208892 | 330 | 9.845e−09 | 7.984e−06 | ADGRB1:12 BNC1:70 VIPR1:74 PRDM16:141 NKX2−4:170 CHRNA4:175 |
| NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON | −0.40857009 | 15 | 4.284e−08 | 2.779e−05 | EP400:1 SFSWAP:9 GOLGA3:231 P2RX2:302 ANKLE2:424 MMP17:427 |
| REACTOME_SIGNALING_BY_GPCR | −0.06452432 | 626 | 4.272e−08 | 2.779e−05 | ADCY1:8 CAMK2B:11 TIAM2:14 TACR2:22 GNAT2:25 NPBWR2:28 |
| MIKKELSEN_MEF_HCP_WITH_H3K27ME3 | −0.06822525 | 553 | 4.803e−08 | 2.833e−05 | CAMK2B:11 ADGRB1:12 CDCP1:40 CNTN4:72 NTSR1:129 SLC15A1:144 |
| REACTOME_DEVELOPMENTAL_BIOLOGY | −0.05329471 | 909 | 6.595e−08 | 3.566e−05 | COL6A1:44 L1CAM:57 RPS6KA2:67 KRT80:76 DSG3:88 COL6A2:89 |
| NIKOLSKY_BREAST_CANCER_7P22_AMPLICON | −0.25221842 | 38 | 7.482e−08 | 3.734e−05 | LFNG:211 INTS1:368 GRIFIN:465 TTYH3:586 ZFAND2A:635 GET4:747 |
| JOHNSTONE_PARVB_TARGETS_3_DN | 0.05622156 | 771 | 1.344e−07 | 6.231e−05 | MDM1:27 MTBP:32 CCPG1:43 FAM13B:68 PPP4R3B:88 MAN2A1:91 |
| DUTERTRE_ESTRADIOL_RESPONSE_24HR_UP | 0.08589001 | 308 | 2.343e−07 | 1.013e−04 | KCNK15:16 NCAPD2:110 ZWILCH:138 STIL:180 MYO19:201 TET2:202 |
| MARSON_BOUND_BY_E2F4_UNSTIMULATED | 0.05943335 | 645 | 3.060e−07 | 1.241e−04 | MDM1:27 MTBP:32 TBCCD1:39 KIAA0825:42 FANCM:50 NCAPD2:110 |
| BLANCO_MELO_BRONCHIAL_EPITHELIAL_CELLS_I | 0.11450159 | 163 | 4.719e−07 | 1.713e−04 | RHNO1:112 KIF24:175 CLSPN:239 ATAD5:270 IRAG2:285 WDHD1:330 |
| REACTOME_INFECTIOUS_DISEASE | −0.05343648 | 779 | 4.752e−07 | 1.713e−04 | SH3GL3:7 ADCY1:8 ACE2:37 TLR8:38 ITPR1:66 DOCK1:90 |
| RODRIGUES_THYROID_CARCINOMA_POORLY_DIFFE | 0.06087136 | 588 | 5.275e−07 | 1.801e−04 | AGGF1:5 LEPROTL1:19 CCPG1:43 SCD:44 PPP4R3B:88 ITCH:98 |
| DACOSTA_UV_RESPONSE_VIA_ERCC3_COMMON_DN | 0.06791195 | 445 | 1.001e−06 | 3.093e−04 | PJA2:7 ZNF292:17 MDN1:21 AKAP9:24 AFF1:35 DST:45 |
| NUYTTEN_EZH2_TARGETS_DN | 0.04783517 | 930 | 9.551e−07 | 3.093e−04 | PJA2:7 LEPROTL1:19 DDHD2:23 MTBP:32 WNK4:51 NBR1:69 |
| DACOSTA_UV_RESPONSE_VIA_ERCC3_DN | 0.05051916 | 815 | 1.150e−06 | 3.393e−04 | SPG11:2 TMEM131L:3 PJA2:7 ZNF292:17 MDN1:21 AKAP9:24 |
| REACTOME_CARDIAC_CONDUCTION | −0.12775290 | 120 | 1.372e−06 | 3.561e−04 | CAMK2B:11 ITPR1:66 KCNJ12:182 SCN1A:219 ATP2B3:298 SCN2A:338 |
| WP_GPCRS_CLASS_A_RHODOPSINLIKE | −0.09662510 | 211 | 1.372e−06 | 3.561e−04 | NPBWR2:28 OPN1LW:63 OPN3:75 NTSR1:129 FFAR3:179 HRH3:271 |
| KAUFFMANN_DNA_REPAIR_GENES | 0.09706905 | 210 | 1.299e−06 | 3.561e−04 | FANCM:50 TEP1:94 FANCE:108 SMC4:118 MBD4:124 MSH3:225 |
| REACTOME_KERATINIZATION | −0.14998450 | 85 | 1.780e−06 | 4.441e−04 | KRT80:76 DSG3:88 KRT13:96 TGM1:109 TGM5:140 KRT5:238 |
| MIKKELSEN_NPC_HCP_WITH_H3K27ME3 | −0.07669899 | 329 | 1.859e−06 | 4.468e−04 | ADGRB1:12 BNC1:70 VIPR1:74 PRDM16:141 CHRNA4:175 CADM3:176 |
| WINTER_HYPOXIA_METAGENE | −0.09239098 | 222 | 2.195e−06 | 5.087e−04 | IGFBP1:41 IGFBP3:43 TGM2:150 SLC6A16:152 NT5E:181 SORL1:198 |
| REACTOME_GPCR_LIGAND_BINDING | −0.06925642 | 398 | 2.304e−06 | 5.154e−04 | TACR2:22 NPBWR2:28 OPN1LW:63 VIPR1:74 OPN3:75 UTS2R:91 |
| BENPORATH_ES_WITH_H3K27ME3 | −0.04464541 | 978 | 2.830e−06 | 6.120e−04 | PLEC:10 CAMK2B:11 VAX2:19 NKPD1:24 FRMPD4:31 NPTX1:32 |
| REACTOME_TRANSMISSION_ACROSS_CHEMICAL_SY | −0.08698466 | 243 | 3.156e−06 | 6.605e−04 | ADCY1:8 CAMK2B:11 GABRR1:52 RPS6KA2:67 GLUL:108 GABRB3:145 |
| GOBERT_OLIGODENDROCYTE_DIFFERENTIATION_U | 0.05855762 | 542 | 3.487e−06 | 7.070e−04 | FBN2:31 CEP192:65 NCAPD2:110 SMC4:118 ZWILCH:138 HAUS6:171 |
| DAZARD_RESPONSE_TO_UV_NHEK_DN | 0.08012816 | 281 | 4.011e−06 | 7.885e−04 | TMEM131L:3 AKAP9:24 AFF1:35 DST:45 SMC4:118 SCAF11:166 |
| WP_STRIATED_MUSCLE_CONTRACTION_PATHWAY | −0.23368138 | 32 | 4.777e−06 | 8.854e−04 | NEB:190 MYBPC2:194 MYH8:261 TNNT3:391 DES:561 TNNT2:672 |
| KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERAC | −0.08253265 | 261 | 4.649e−06 | 8.854e−04 | TACR2:22 NPBWR2:28 GABRR1:52 VIPR1:74 UTS2R:91 NTSR1:129 |
| REACTOME_NEUROTRANSMITTER_RECEPTORS_AND_ | −0.09791101 | 182 | 5.394e−06 | 9.457e−04 | ADCY1:8 CAMK2B:11 GABRR1:52 RPS6KA2:67 GABRB3:145 CHRNA4:175 |
| REACTOME_FORMATION_OF_THE_CORNIFIED_ENVE | −0.14887774 | 78 | 5.539e−06 | 9.457e−04 | KRT80:76 DSG3:88 KRT13:96 TGM1:109 TGM5:140 KRT5:238 |
| RODRIGUES_THYROID_CARCINOMA_ANAPLASTIC_U | 0.05352207 | 627 | 5.405e−06 | 9.457e−04 | WDR36:74 PPP4R3B:88 ITCH:98 SMC4:118 ZWILCH:138 HAUS6:171 |
| REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY | −0.05395310 | 607 | 6.352e−06 | 1.057e−03 | CAMK2B:11 TRIM29:18 IL37:39 TNFRSF4:46 RPS6KA2:67 P4HB:68 |
| RICKMAN_METASTASIS_DN | −0.08647918 | 227 | 7.447e−06 | 1.208e−03 | TRIM29:18 ITGB4:26 CDCP1:40 RNH1:49 DSG3:88 IL18:107 |

# DisGeNET Top pathways by non−permulation

| Geneset | stat | num.genes | pval | p.adj | gene.vals |
|---|---|---|---|---|---|
| HIV Infections | −0.06391070 | 626 | 6.598e−08 | 6.473e−04 | PLEC:10 TLR8:38 VIPR1:74 APOA1:84 ISG20:101 IL18:107 |
| Primary microcephaly | 0.12933094 | 104 | 5.362e−06 | 2.630e−02 | FANCM:50 FANCE:108 STIL:180 CDK5RAP2:200 TRMT10A:259 MFSD2A:265 |
| Congenital cerebral hernia | 0.18408151 | 46 | 1.582e−05 | 5.172e−02 | CRB1:287 WDPCP:289 RDH12:387 ARHGAP31:476 KCNJ13:897 FLNB:927 |
| CATARACT, AUTOSOMAL DOMINANT | −0.37959311 | 10 | 3.232e−05 | 5.285e−02 | CRYAA2:47 GJA8:54 CRYAA:118 CRYBB2:221 BFSP2:447 MIP:466 |
| Nuclear cataract | −0.21448307 | 32 | 2.698e−05 | 5.285e−02 | HSF4:36 GJA8:54 CRYAA:118 TGM2:150 CRYBB2:221 FTO:380 |
| Virus Diseases | −0.04780109 | 678 | 2.721e−05 | 5.285e−02 | ADGRB1:12 LEXM:29 ACE2:37 TLR8:38 TNFRSF4:46 OPN1LW:63 |
| Mouth Neoplasms | −0.11577652 | 106 | 3.925e−05 | 5.500e−02 | UPP1:15 HMGCS2:50 IL18:107 PTK2:165 EGFR:212 CAV1:468 |
| Autism Spectrum Disorders | −0.05362741 | 481 | 6.421e−05 | 7.874e−02 | SFSWAP:9 SHANK2:13 L1CAM:57 HERC2:58 JAKMIP1:60 ITPR1:66 |
| Cataract, Central Saccular, With Sutural | −0.46255736 | 6 | 8.714e−05 | 7.939e−02 | GJA8:54 CRYBB2:221 BFSP2:447 MIP:466 CRYBA1:777 CRYGS:1458 |
| Night blindness, congenital stationary | −0.22853271 | 25 | 7.677e−05 | 7.939e−02 | TRPM1:17 PDE6B:48 CACNA2D4:167 GRK7:178 NYX:196 GRM6:268 |
| Asthma | −0.03632998 | 1057 | 8.902e−05 | 7.939e−02 | MMEL1:5 SFSWAP:9 IGSF1:16 TACR2:22 ITGB4:26 TLR8:38 |
| Complete congenital stationary night bli | −0.30084589 | 14 | 9.734e−05 | 7.958e−02 | TRPM1:17 PDE6B:48 NYX:196 GRM6:268 FGFR2:645 CACNA1F:1166 |
| Anemia, hereditary spherocytic hemolytic | 0.49056707 | 5 | 1.453e−04 | 8.027e−02 | SPTA1:47 ANK1:53 SLC4A1:195 EPB42:235 SPTB:244 NA |
| Cardiomyopathies | −0.05470512 | 420 | 1.339e−04 | 8.027e−02 | PLEC:10 ACE2:37 UTS2R:91 MYH2:100 ISG20:101 IL18:107 |
| Nasopharyngeal carcinoma | −0.04593978 | 603 | 1.365e−04 | 8.027e−02 | IGFBP3:43 OPN1LW:63 TRAF2:80 IL18:107 CD40:110 IL16:134 |
| Oguchi disease | −0.27810641 | 16 | 1.176e−04 | 8.027e−02 | TRPM1:17 PDE6B:48 GRK7:178 NYX:196 GRM6:268 FGFR2:645 |
| Sloping forehead | 0.12366245 | 79 | 1.473e−04 | 8.027e−02 | STIL:180 CDK5RAP2:200 MFSD2A:265 WDPCP:289 ERCC6:296 ATR:347 |
| Spherocytosis | 0.49056707 | 5 | 1.453e−04 | 8.027e−02 | SPTA1:47 ANK1:53 SLC4A1:195 EPB42:235 SPTB:244 NA |
| Lymphoma | −0.03478731 | 1065 | 1.664e−04 | 8.592e−02 | ADGRG7:4 OPN1LW:63 TRAF2:80 ISG20:101 IL18:107 CD40:110 |
| NIGHT BLINDNESS, CONGENITAL STATIONARY, | −0.30860985 | 12 | 2.144e−04 | 8.795e−02 | TRPM1:17 PDE6B:48 NYX:196 GRM6:268 CACNA1F:1166 CABP4:1637 |
| Arthritis | −0.04754330 | 516 | 2.476e−04 | 8.795e−02 | TLR8:38 IL37:39 IGFBP3:43 TRAF2:80 APOA1:84 CLEC2A:93 |
| Ciliopathies | 0.08160786 | 170 | 2.510e−04 | 8.795e−02 | ARL13B:55 INVS:106 PKHD1:223 WDPCP:289 NPHP3:297 ATR:347 |
| Epidermolysis Bullosa Simplex | −0.23342433 | 21 | 2.136e−04 | 8.795e−02 | PLEC:10 ITGB4:26 KRT80:76 TGM5:140 KRT5:238 DES:561 |
| Hypochloremia (disorder) | 0.43162025 | 6 | 2.508e−04 | 8.795e−02 | CLCNKB:15 CLCNKA:38 KCNJ1:617 SLC12A1:1480 BSND:2058 SLC26A3:2380 |
| Nuclear non−senile cataract | −0.20035453 | 29 | 1.891e−04 | 8.795e−02 | GJA8:54 CRYAA:118 CRYBB2:221 FTO:380 MIP:466 CRYBA2:493 |
| Serum chloride level decreased (finding) | 0.43162025 | 6 | 2.508e−04 | 8.795e−02 | CLCNKB:15 CLCNKA:38 KCNJ1:617 SLC12A1:1480 BSND:2058 SLC26A3:2380 |
| Tetany | 0.23137323 | 21 | 2.428e−04 | 8.795e−02 | TRPM6:34 CLDN16:41 JMJD1C:163 KCNJ1:617 GCM2:1037 SLC12A3:1100 |
| Wolf−Hirschhorn Syndrome | −0.28280984 | 14 | 2.488e−04 | 8.795e−02 | SLBP:116 NSD2:243 LETM1:1030 CPLX1:1181 LAP3:1451 NELFA:1997 |
| CATARACT, COPPOCK−LIKE | −0.33002739 | 10 | 3.018e−04 | 8.973e−02 | GJA8:54 ISG20:101 CRYBB2:221 BFSP2:447 VIM:801 GJA3:915 |
| Cataract, Pulverulent | −0.33048122 | 10 | 2.961e−04 | 8.973e−02 | HSF4:36 GJA8:54 CRYBB2:221 BFSP2:447 VIM:801 GJA3:915 |
| Renal salt wasting | 0.26232729 | 16 | 2.807e−04 | 8.973e−02 | CLCNKB:15 CLCNKA:38 CYP11A1:159 STAR:319 KCNJ1:617 SARS2:911 |
| Schizophrenia | −0.02789078 | 1613 | 2.864e−04 | 8.973e−02 | ADCY1:8 SFSWAP:9 PLEC:10 CAMK2B:11 SHANK2:13 TRPM1:17 |
| Severe myopia | −0.11038898 | 91 | 2.783e−04 | 8.973e−02 | TRPM1:17 IGFBP3:43 PDE6B:48 OPN1LW:63 CACNA2D4:167 NYX:196 |
| Chronic Lymphocytic Leukemia | −0.03660776 | 866 | 3.144e−04 | 9.071e−02 | RNH1:49 OPN1LW:63 ITPR1:66 TRAF2:80 MYH2:100 ISG20:101 |
| Skin Diseases, Genetic | −0.17244808 | 36 | 3.448e−04 | 9.649e−02 | ITGB4:26 IGFBP3:43 TGM1:109 KRT5:238 ITGA6:405 FLCN:547 |
| Nephritis | −0.09268977 | 125 | 3.541e−04 | 9.649e−02 | ACE2:37 IL18:107 CD40:110 PTK2:165 C3:249 TGFB1:324 |
| Familial intrahepatic cholestasis of pre | 0.41652326 | 6 | 4.103e−04 | 1.035e−01 | ATP8B1:122 ABCB4:207 ABCB11:381 NR1H4:806 VCAM1:2269 TNF:4274 |
| Genu recurvatum | −0.27269881 | 14 | 4.116e−04 | 1.035e−01 | CANT1:326 COL1A2:407 COL5A1:432 CTDP1:857 COL5A2:904 KRAS:1594 |
| Granulomatosis with polyangiitis | −0.14068210 | 53 | 3.996e−04 | 1.035e−01 | TNFRSF4:46 DPP4:161 TGFB1:324 MIP:466 MMP2:505 SERPINA5:706 |
| Waddling gait | −0.14716587 | 48 | 4.231e−04 | 1.038e−01 | L1CAM:57 NEB:190 BICD2:203 BIN1:272 TGFB1:324 CANT1:326 |

# MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

| Geneset | stat | num.genes | pval | p.adj | gene.vals |
|---|---|---|---|---|---|
| MP0003635 abnormal synaptic transmissio | −0.06803867 | 423 | 3.307e−06 | 1.574e−03 | ADCY1:8 CAMK2B:11 L1CAM:57 ITPR1:66 AFF2:73 NTSR1:129 |
| MP0005365 abnormal bile salt | 0.23750816 | 26 | 2.857e−05 | 3.638e−03 | ATP8B1:122 PNLIP:198 EHHADH:348 ABCB11:381 CYP7A1:385 ALB:416 |
| MP0001970 abnormal pain threshold | −0.09969054 | 150 | 3.057e−05 | 3.638e−03 | ADCY1:8 GABRR1:52 AFF2:73 TSHZ3:95 NTSR1:129 PTGDS:153 |
| MP0003252 abnormal bile duct | 0.25935804 | 22 | 2.611e−05 | 3.638e−03 | ATP8B1:122 ABCB4:207 PKHD1:223 ABCB11:381 NR1H4:806 ABCG5:886 |
| MP0002063 abnormal learning/memory/cond | −0.05698131 | 353 | 3.392e−04 | 3.226e−02 | ADCY1:8 L1CAM:57 AFF2:73 RAI1:112 ARC:149 SLC18A2:159 |
| MP0009745 abnormal behavioral response | −0.07287687 | 204 | 4.066e−04 | 3.226e−02 | UPP1:15 GABRB3:145 SLC18A2:159 CHRNA4:175 GAL:213 SCN1A:219 |
| MP0003868 abnormal feces composition | 0.18434602 | 25 | 1.447e−03 | 8.941e−02 | PNLIP:198 ABCB4:207 APC:279 CYP7A1:385 ALB:416 ITGB1:959 |
| MP0004019 abnormal vitamin homeostasis | 0.14366960 | 41 | 1.503e−03 | 8.941e−02 | CLDN16:41 PLIN2:76 KCP:156 TGIF1:272 CYP7A1:385 SLC23A1:427 |
| MP0000598 abnormal liver morphology | 0.04367209 | 469 | 1.757e−03 | 9.257e−02 | ANK1:53 ANGPTL3:72 MAN2A1:91 INVS:106 ATP8B1:122 CYP1A1:186 |
| MP0005085 abnormal gallbladder physiolo | 0.17765172 | 25 | 2.144e−03 | 9.257e−02 | ATP8B1:122 ABCB4:207 ABCB11:381 CYP7A1:385 NR1H4:806 ABCG5:886 |
| MP0000678 abnormal parathyroid gland | 0.19063313 | 21 | 2.528e−03 | 9.257e−02 | CHRD:82 GCM2:1037 PTHLH:1144 CYP27B1:1538 PTH:1900 TBX1:2000 |
| MP0002272 abnormal nervous system | −0.06365574 | 195 | 2.510e−03 | 9.257e−02 | PLEC:10 GABRR1:52 RAI1:112 GABRB3:145 ARC:149 CHRNA4:175 |
| MP0010094 abnormal chromosome stability | 0.11335117 | 61 | 2.297e−03 | 9.257e−02 | USP1:273 ATR:347 PARP2:589 CENPE:612 BUB1B:632 AICDA:652 |
| MP0002572 abnormal emotion/affect behav | −0.05202699 | 289 | 2.888e−03 | 9.821e−02 | ADCY1:8 L1CAM:57 RAI1:112 GABRB3:145 ARC:149 SLC18A2:159 |
| MP0000609 abnormal liver physiology | 0.04468041 | 373 | 3.945e−03 | 1.191e−01 | ANGPTL3:72 PLIN2:76 MAN2A1:91 INVS:106 ATP8B1:122 CPT1C:153 |
| MP0001666 abnormal nutrient absorption | 0.12871503 | 42 | 4.002e−03 | 1.191e−01 | PNLIP:198 ABCB4:207 SOAT2:229 CYP7A1:385 ALB:416 SLC23A1:427 |
| MP0002135 abnormal kidney morphology | 0.03931902 | 449 | 5.751e−03 | 1.610e−01 | ANK1:53 MAN2A1:91 INVS:106 KCP:156 SLC4A1:195 TET2:202 |
| MP0005636 abnormal mineral homeostasis | 0.06921186 | 134 | 6.143e−03 | 1.625e−01 | TRPM6:34 CLDN16:41 WNK4:51 ANK1:53 CYP11A1:159 SLC4A1:195 |
| MP0002136 abnormal kidney physiology | 0.04798951 | 278 | 6.970e−03 | 1.659e−01 | SLC14A2:9 CLCNKA:38 CLDN16:41 WNK4:51 MAN2A1:91 SLC4A1:195 |
| MP0002064 seizures | −0.05048566 | 252 | 6.737e−03 | 1.659e−01 | ITPR1:66 DBNL:81 RAI1:112 GABRB3:145 CHRNA4:175 GAL:213 |
| MP0008873 increased physiological sensi | −0.06292711 | 153 | 7.881e−03 | 1.786e−01 | TGM2:150 SLC18A2:159 PTK2:165 GAL:213 RAE1:236 BIN1:272 |
| MP0002733 abnormal thermal nociception | −0.07859221 | 92 | 9.640e−03 | 1.942e−01 | L1CAM:57 AFF2:73 NTSR1:129 CACNA1B:265 SLC6A2:285 HOXB8:613 |
| MP0005551 abnormal eye electrophysiolog | −0.06376925 | 139 | 1.017e−02 | 1.942e−01 | TRPM1:17 GNAT2:25 NPTX1:32 PDE6B:48 GABRR1:52 GJA8:54 |
| MP0005253 abnormal eye physiology | −0.07884946 | 90 | 1.020e−02 | 1.942e−01 | TRPM1:17 GJA8:54 RGS9BP:184 BFSP2:447 MIP:466 NR2E3:475 |
| MP0002148 abnormal hypersensitivity rea | −0.07437968 | 102 | 9.967e−03 | 1.942e−01 | TNFRSF4:46 CD40:110 PTGER4:286 CD81:414 TBXA2R:559 ANXA5:615 |
| MP0002085 abnormal embryonic tissue | 0.03091313 | 645 | 1.079e−02 | 1.975e−01 | ENPP2:30 TRPM6:34 ARL13B:55 CHRD:82 ECSIT:87 HESX1:119 |
| MP0003453 abnormal keratinocyte physiol | −0.09550606 | 58 | 1.222e−02 | 2.155e−01 | ITGB4:26 PTK2:165 DSC3:364 PKP3:441 FGFR2:645 KRT14:788 |
| MP0002139 abnormal hepatobiliary system | 0.24909590 | 8 | 1.475e−02 | 2.259e−01 | ABCB4:207 ALDH1A1:935 FABP1:1214 ABCG8:2004 NR1H3:2492 INSIG1:9066.5 |
| MP0009643 abnormal urine homeostasis | 0.04774471 | 226 | 1.498e−02 | 2.259e−01 | SLC14A2:9 CLCNKA:38 CLDN16:41 MAN2A1:91 SLC4A1:195 ARSB:242 |
| MP0002084 abnormal developmental patter | 0.03892572 | 355 | 1.411e−02 | 2.259e−01 | ENPP2:30 ARL13B:55 CHRD:82 ECSIT:87 INVS:106 LAMA1:149 |
| MP0010771 integument phenotype | −0.05258815 | 184 | 1.519e−02 | 2.259e−01 | PLEC:10 ITGB4:26 TGM1:109 PTK2:165 HSF1:171 EGFR:212 |
| MP0002067 abnormal sensory capabilities | −0.04457094 | 260 | 1.519e−02 | 2.259e−01 | ITPR1:66 DBNL:81 RAI1:112 GABRB3:145 SCN1A:219 CACNA1B:265 |
| MP0008058 abnormal DNA repair | 0.11202928 | 39 | 1.576e−02 | 2.273e−01 | TEX15:236 APC:279 ERCC6:296 PARP2:589 POLK:943 BRCA2:1104 |
| MP0002734 abnormal mechanical nocicepti | −0.08275632 | 69 | 1.800e−02 | 2.520e−01 | ADCY1:8 GABRR1:52 L1CAM:57 TSHZ3:95 CACNA1B:265 STOML3:581 |
| MP0001177 atelectasis | −0.09245748 | 54 | 1.921e−02 | 2.613e−01 | DOCK1:90 TSHZ3:95 EGFR:212 KCNQ2:327 PDGFA:451 SLC5A3:570 |
| MP0002970 abnormal white adipose | −0.05327761 | 162 | 2.073e−02 | 2.741e−01 | RPTOR:53 IL18:107 RAI1:112 NTSR1:129 EGFR:212 C3:249 |
| MP0005332 abnormal amino acid | 0.06018448 | 116 | 2.637e−02 | 3.304e−01 | ARG1:292 GCNT2:316 SLC6A6:357 LEPR:417 ACADM:530 GNMT:540 |
| MP0002876 abnormal thyroid physiology | 0.15173997 | 18 | 2.601e−02 | 3.304e−01 | ABCA5:958 GLIS3:966 CGA:1401 LEP:1515 THRB:3544 HPN:3650 |
| MP0003195 calcinosis | 0.22298281 | 8 | 2.904e−02 | 3.544e−01 | ENPP1:999 SLC12A1:1480 AGXT:2118 ABCC6:2384 CASR:3577 SOD2:9066.5 |
| MP0000313 abnormal cell death | 0.03041140 | 452 | 3.215e−02 | 3.826e−01 | ENPP2:30 PARG:46 MBD4:124 LAMA1:149 STIL:180 APC:279 |

# tissue_specific Top pathways by non−permulation

| Geneset | stat | num.genes | pval | p.adj | gene.vals |
|---|---|---|---|---|---|
| EBV.lymphocyte | 0.05303264 | 669 | 6.110e−06 | 2.842e−04 | TMEM131L:3 MTBP:32 NCAPD2:110 SMC4:118 IL23R:146 STIL:180 |
| liver | 0.07191216 | 328 | 1.052e−05 | 2.842e−04 | ACOX2:63 ANGPTL3:72 MOGAT1:84 SERPINA7:100 DMGDH:113 ACOT12:164 |
| pancreas | 0.12426695 | 94 | 3.388e−05 | 6.099e−04 | CPB1:6 TMED6:52 CTRC:60 KCNK16:134 PNLIP:198 RBPJL:361 |
| cerebral.cortex | −0.11015734 | 107 | 8.958e−05 | 1.209e−03 | ADGRB1:12 MGAT5B:206 C1QL2:388 MAS1:449 FAM163B:464 RCVRN:511 |
| leg.skin | −0.07515441 | 217 | 1.594e−04 | 1.722e−03 | TRPM1:17 TRIM29:18 VSIG8:20 NKPD1:24 KLK7:35 IL37:39 |
| Brodmann.area.9 | −0.07700150 | 182 | 3.823e−04 | 2.950e−03 | FRMPD4:31 TMEM132D:111 GABRB3:145 SCN1A:219 NETO1:267 KCNQ3:435 |
| esophagus.mucosa | −0.07094612 | 220 | 3.329e−04 | 2.950e−03 | LEXM:29 CDCP1:40 BNC1:70 DSG3:88 KRT13:96 CAPN14:97 |
| suprapubic.skin | −0.07123585 | 193 | 7.218e−04 | 4.872e−03 | NKPD1:24 IL37:39 PARD6G:65 BNC1:70 KRT80:76 DSG3:88 |
| skeletal.muscle | −0.05576449 | 238 | 3.393e−03 | 2.036e−02 | IGFN1:55 MYH2:100 RGS9BP:184 NEB:190 MYBPC2:194 KLHL30:226 |
| vagina | −0.08790947 | 90 | 4.099e−03 | 2.213e−02 | LEXM:29 BNC1:70 DSG3:88 KRT13:96 SLC18A2:159 KRT5:238 |
| cerebellum | −0.03585322 | 374 | 1.940e−02 | 9.523e−02 | RNF112:3 L1CAM:57 ZBTB46:105 MYT1:146 NOS1AP:177 GLRA4:180 |
| thyroid | 0.04848450 | 150 | 4.184e−02 | 1.883e−01 | KCNJ16:37 CLCNKA:38 PRTG:71 WDR72:116 PLA2R1:126 RASEF:161 |
| cortex.kidney | 0.05453275 | 114 | 4.543e−02 | 1.887e−01 | CLCNKB:15 CLDN16:41 TINAG:154 SLC5A12:172 PKHD1:223 TMEM52B:551 |
| ectocervix | 0.10140984 | 25 | 7.956e−02 | 2.864e−01 | RIPOR3:384 COL27A1:851 PTHLH:1144 PAPPA2:1467 ADRA2A:2001 FGF7:2918 |
| fallopian.tube | 0.07081991 | 52 | 7.799e−02 | 2.864e−01 | ADAMTS3:97 GREB1L:372 GMNC:391 CCDC17:532 GAS2L2:808 ATP8B4:1137 |
| sigmoid.colon | −0.06644878 | 50 | 1.048e−01 | 3.538e−01 | TACR2:22 NTSR1:129 LIMS2:503 DES:561 GREM2:702 CNN1:936 |
| prostate | −0.07254683 | 38 | 1.224e−01 | 3.887e−01 | KRT5:238 CDH26:277 P2RX2:302 NR2E3:475 ACP3:989 SRD5A2:1184 |
| adrenal.gland | 0.04003804 | 108 | 1.526e−01 | 4.579e−01 | SLC14A2:9 ADGRV1:12 CYP17A1:33 CYP11A1:159 CELA1:168 STAR:319 |
| hypothalamus | −0.04522573 | 66 | 2.052e−01 | 5.324e−01 | GABRG3:483 PIRT:822 MC4R:830 SYT6:838 SMIM17:1005 GRIK1:1197 |
| testis | −0.01093795 | 1380 | 2.063e−01 | 5.324e−01 | MMEL1:5 SH3GL3:7 LEXM:29 ACE2:37 IL37:39 EPS8L3:51 |
| tibial.artery | −0.03548029 | 107 | 2.071e−01 | 5.324e−01 | SDK1:85 C1QTNF1:283 CDH6:373 ADARB1:381 LMOD1:431 WFDC1:516 |
| spleen | 0.02103712 | 298 | 2.182e−01 | 5.356e−01 | PATL2:58 AKNA:150 STAB2:165 BANK1:185 SOAT2:229 IL24:310 |
| minor.salivary.gland | 0.04206065 | 69 | 2.285e−01 | 5.364e−01 | ENPP3:61 EHF:794 ASCL3:1403 IRX1:1420 TCN1:1606 BPIFB1:1610 |
| aorta | 0.02537214 | 154 | 2.807e−01 | 5.413e−01 | KCNK15:16 HMCN1:20 SUSD5:252 VCAN:256 PTGIS:614 PCDH11X:636 |
| caudate.nucleus | −0.06383683 | 26 | 2.604e−01 | 5.413e−01 | TAC1:808 VAX1:1226 RLBP1:1277 GPR88:1394 MLC1:1449 PDE1B:1480 |
| cerebellar.hemisphere | −0.01459895 | 502 | 2.747e−01 | 5.413e−01 | RNF112:3 ADCY1:8 CAMK2B:11 VAX2:19 NPTX1:32 AFF2:73 |
| lung | 0.02786056 | 139 | 2.599e−01 | 5.413e−01 | RTKN2:290 GP9:528 HTR3C:670 ROS1:698 PTPRB:812 ADGRF5:1185 |
| subcutaneous.adipose | 0.03791947 | 72 | 2.675e−01 | 5.413e−01 | PLIN2:76 CYP1A1:186 LVRN:404 PPARG:542 FAM124B:607 PFKFB3:631 |
| atrium.auricle | 0.02898090 | 90 | 3.440e−01 | 5.550e−01 | CCDC141:268 MLIP:294 TNNI3K:349 MYPN:440 PPP1R3A:553 MYBPC3:700 |
| Brodmann.area.24 | −0.04972075 | 31 | 3.387e−01 | 5.550e−01 | FAM189A1:197 STUM:515 DRD5:797 KCNS1:869 NEUROD6:983 TMEM271:2418 |
| coronary.artery | 0.07226993 | 15 | 3.328e−01 | 5.550e−01 | SUSD5:252 SOST:1574 MYO18B:1789 GJA5:3105 CCN3:3479 MTHFD1L:3595 |
| hippocampus.proper | −0.08367333 | 10 | 3.597e−01 | 5.550e−01 | NEUROD6:983 OLIG2:2015 NEUROD2:6960.5 SLC17A7:6960.5 HRK:6960.5 CABP7:6960.5 |
| peyers.patch | 0.02270987 | 167 | 3.151e−01 | 5.550e−01 | SLC6A20:123 TINAG:154 SLC5A12:172 MYRFL:215 SLC23A1:427 CREB3L3:547 |
| putamen | −0.06903656 | 15 | 3.549e−01 | 5.550e−01 | DIPK1C:473 PDE10A:854 GPR88:1394 LRRC10B:2174 RGS9:2289 ANO3:6960.5 |
| transverse.colon | 0.02412505 | 150 | 3.112e−01 | 5.550e−01 | TRPM6:34 WNK4:51 ATP8B1:122 TINAG:154 DQX1:669 SLC44A4:728 |
| left.ventricle | −0.03139967 | 68 | 3.721e−01 | 5.581e−01 | MYH7B:132 LMOD2:252 MYOM2:425 FSD2:485 CKMT2:500 HSPB3:502 |
| uterus | 0.03240295 | 60 | 3.866e−01 | 5.643e−01 | CHRDL2:179 TEX15:236 WT1:571 CLEC4F:888 ENPP1:999 MEIS3:1317 |
| bladder | 0.04499107 | 24 | 4.460e−01 | 6.136e−01 | SNX31:488 UPK3A:844 DHRS2:2647 CLEC3A:3230 PADI3:3629 SEMA3E:3997 |
| blood | 0.01327860 | 275 | 4.545e−01 | 6.136e−01 | TREML2:89 DHX34:132 LY6G6F:145 SLC4A1:195 RHAG:227 EPB42:235 |
| nucleus.accumbens | −0.02274130 | 99 | 4.364e−01 | 6.136e−01 | JAKMIP1:60 INSYN2A:123 PRKCG:257 OPRK1:534 GABRA5:606 CDH4:709 |